# Reminders SaaS Application

## 1 . Purpose

A date management system designed to manage the deadlines of items required for operat company (such as permits, contracts, payments, inspections, and training). The system centrall manages these items together with related documents in order to prevent missed renewals an required actions.

---

# 2 . Core Concept

This application manages various operational entities of a company along a timeline.

Each entity is associated with time-related information such as issue dates, execution dates, expiration dates. By comparing these dates with the current date, the system determines wh renewal is required, whether actions have been missed, or whether an item has expired.

Reminders and calendar views are simply ways to display these results. The fundamental purpo system is to manage the relationship between entities and time.

---

# 3 . Input

## 1 . Item

Entities within company operations that have deadlines or required action dates.

Examples:

- Liquor License
- Insurance Policy
- Health Inspection
- Food Safety Training
- Hood Cleaning
- Tax Payment

---

## 2 . Reference Date

The date when the entity occurred, was obtained, or was executed.

Examples:

- License issued date
- Training completion date
- Inspection date
- Contract start date

---

### 3 . Due Date

The date until which the entity is valid, or the date by which an action must be completed.

Examples:

- License expiration date
- Tax due date
- Next inspection date

---

# 4 . Processing Logic

## Main Processing Flow

### 1 . Retrieve Date Information

The system retrieves date information associated with each entity, such as reference dates, due scheduled dates.

---

### 2 . Compare with the Current Date

The system compares the current date with the entity's relevant date.

Example:

```
today < due_date  → upcoming
today = due_date  → due today
today > due_date  → overdue
```

---

### 3 . Determine Status

Based on the comparison results, the system determines the current status of the entity.

Examples:

- upcoming
- due today
- overdue
- completed

---

### Additional Processing

4 **. Calculate the Next Date**

If the entity is configured as a repeating item, the system calculates the next due date or according to predefined rules.

5 **. Determine Notification Timing**

The system generates reminder notifications at predefined times relative to the due date or date.

6 **. Generate the Next Schedule**

If required, the system creates the next due date or scheduled item as a new record.

---

# 5 . Outputs

## Main Output Information

### 1 . Entity Information

What the item is (e.g., license, training, inspection).

### 2 . Relevant Dates

Due dates, scheduled dates, or next occurrence dates.

### 3 . Current Status

Examples:

- upcoming
- due today
- overdue
- completed

### 4 . Required Actions

Examples:

- renewal
- submission
- execution
- confirmation

---

# 6 . Configuration / Rules

This section defines the rules and parameters used by the system during processing.

The Processing Logic refers to these rules to perform calculations, determine statuses, and ge
future schedules.

---

## Reminder Sequence

Defines when reminder notifications should be sent relative to a due date or scheduled date.
notifications to be sent shortly before a deadline or a defined period in advance.

---

## Repeat

Defines whether the entity occurs once or repeatedly over time.

- yes
- no

---

## Repeat Interval

If repetition is required, this defines the interval at which the next due date or scheduled
generated.

Examples:

- every 1 year
- every 3 months
- every month

---

## Repeat Trigger

Defines when the next record or scheduled item should be generated.

- **on completion**
  The next item is created when the current entity is completed.

- **scheduled date**
  The next item is created based on a predefined date regardless of completion status.

---

## Generation Horizon

Defines how far into the future repeating schedules should be generated. This ensures that schedules are visible in calendars while preventing unlimited record generation.

---

## Expiry Rule

Defines how expiration dates are calculated. These rules support expiration systems that canno expressed with simple day-based calculations.

- **Anniversary-based**
  The expiration date is calculated based on a fixed period after the issue or completion date.

- **Calendar-year-based**
  The expiration date is determined based on the calendar year (e.g., valid until Decembe of that year).

- **Fiscal-year-based**
  The expiration date is determined based on the fiscal or administrative year.

- **Month-end-based**
  The expiration date is rounded to the end of the month.

- **Fixed-date**
  The expiration date occurs on a specific date each year.